E-Commerce Sales Data Analysis Using SQL

I used **Microsoft SQL Server** to create and analyse an e-commerce sales database. The database consisted of multiple relational tables, including Customers, Products, Orders, OrderItems, and Payments tables to extract meaningful business insights.

I performed a join between **Orders and Customers** to link each order with customer-level information such as name and registration details. I then joined **Payments** with the Orders table to combine payment-related details, including payment method and amount. This combined dataset allowed me to analyse complete transaction-level information in a single view.

Using this unified data, I calculated the **total sales generated by each customer** and sorted the results in **descending order** to identify the **top 5 highest-spending customers**. I also computed the **Average Order Value (AOV)** to measure the typical revenue per order. Additionally, I analysed the **payment method breakdown** to understand which payment modes (UPI, Card, Wallet, etc.) were most frequently used by customers.

Overall, this SQL analysis provides clear insights into customer spending patterns, top-value customers, and payment preferences. These insights are useful for understanding business performance and supporting data-driven decision-making.

Database Schema:

- List of tables: Customers, Orders, Products, OrderItems, Payments

SQL Queries:

```
use master

go

create database Ecommerce

go

sp_helpdb 'Ecommerce'

go
```

```
use Ecommerce
go

--creating customer table
create table Customer_Details
(
 customerID                INT                          IDENTITY(1,1) PRIMARY KEY,,
 FirstName                 VARCHAR(50),
 LastName                  VARCHAR(50),
 EmailAddress              VARCHAR(100),
 RegistrationDate          DATE,
 Address                   VARCHAR(150)
)
go

--Inserting sample customer data

insert into Customer_Details( FirstName,LastName,EmailAddress,RegistrationDate,Address)
values
('Shiva', 'Kranthi', 'shiva.kranthi@example.com', '2024-01-05', 'Madhapur, Hyderabad'),
('Umesh', 'Reddy', 'umesh.reddy@example.com', '2024-02-10', 'Kukatpally, Hyderabad'),
('Kranthi', 'Kumar', 'kranti.kumar@example.com', '2024-03-12', 'Ameerpet, Hyderabad'),
('Sai', 'Kiran', 'sai.kiran@example.com', '2024-04-15', 'Gachibowli, Hyderabad'),
('Harshitha', 'Devi', 'harshitha.devi@example.com', '2024-05-20', 'Manikonda, Hyderabad'),
('Shivani', 'Rao', 'shivani.rao@example.com', '2024-06-18', 'Secunderabad, Hyderabad'),
```

```
('Sravanthi', 'Reddy', 'sravanthi.reddy@example.com', '2024-07-25', 'Kondapur, Hyderabad'),
('Chiranjeevi', 'Varma', 'chiru.varma@example.com', '2024-08-04', 'Lingampally, Hyderabad'),
('Charan', 'Teja', 'charan.teja@example.com', '2024-09-09', 'Banjara Hills, Hyderabad'),
('Thirumal', 'Netha', 'thirumal.netha@example.com', '2024-10-14', 'Miyapur, Hyderabad'),
('Anitha', 'Kumari', 'anitha.kumari@example.com', '2024-11-02', 'Hitech City, Hyderabad');

--To find the total number of customers
SELECT 'Customer_Details' AS TableName, COUNT(*) AS Rows FROM Customer_Details;

select*from Customer_Details
go

--Creating products table
CREATE TABLE Products (
   ProductID          INT                          IDENTITY(1,1) PRIMARY KEY,
   ProductName        VARCHAR(100),
   Category           VARCHAR(50),
   Price              DECIMAL(10,2),
   Stock              INT
)
go

INSERT INTO Products (ProductName, Category, Price, Stock)
VALUES
('Bluetooth Earbuds', 'Electronics', 1499.00, 120),
```

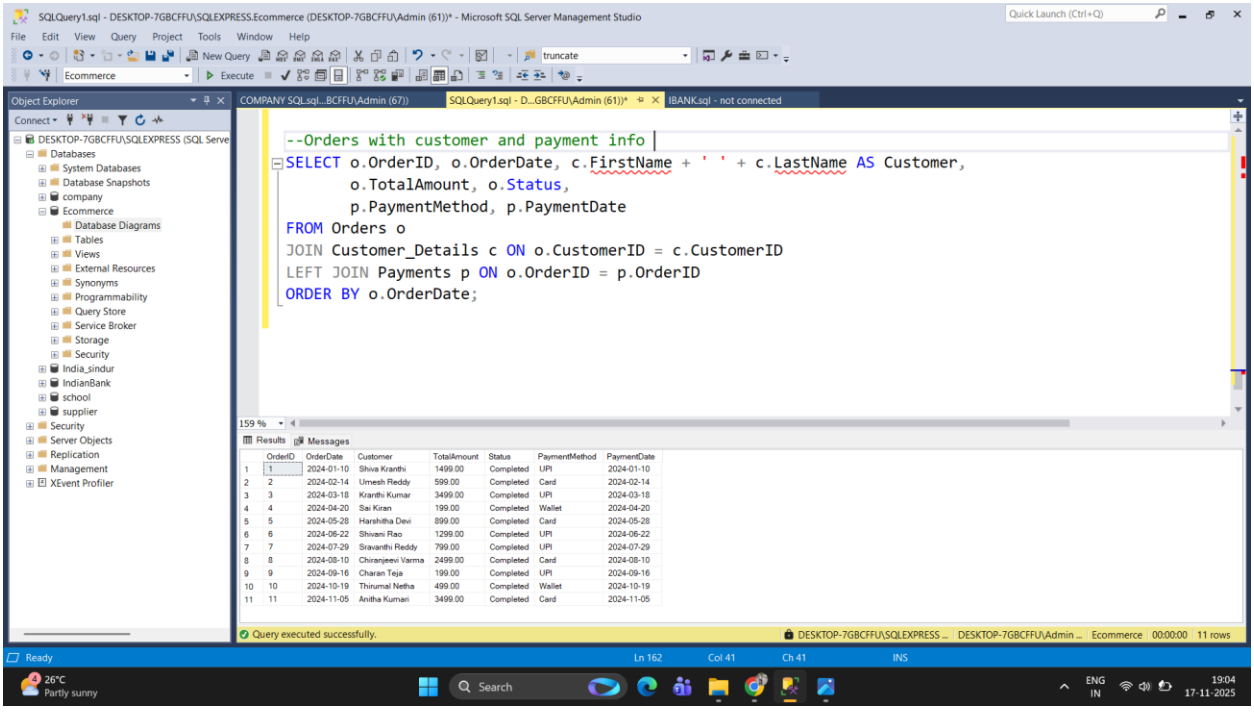
```
('Wireless Mouse', 'Electronics', 599.00, 200),
('Office Chair', 'Furniture', 3499.00, 50),
('Notebook Set', 'Stationery', 199.00, 500),
('Water Bottle', 'Lifestyle', 299.00, 300),
('Backpack', 'Lifestyle', 899.00, 150),
('Power Bank 10,000mAh', 'Electronics', 1299.00, 100),
('LED Table Lamp', 'Home Decor', 799.00, 80),
('Smart Watch', 'Electronics', 2499.00, 70),
('USB-C Cable', 'Accessories', 199.00, 400),
('Yoga Mat', 'Sports', 499.00, 90)

--To find the Total number of products
SELECT 'Products'  AS TableName, COUNT(*) AS Rows FROM Products;

select*from Products
go

CREATE TABLE Orders (
   OrderID                INT IDENTITY(1,1)        PRIMARY KEY,
   CustomerID             INT      FOREIGN KEY REFERENCES Customer_Details (CustomerID),
   OrderDate              DATE,
   TotalAmount            DECIMAL(10,2),
   Status                 VARCHAR(20)

)
```

```
go


INSERT INTO Orders (CustomerID, OrderDate, TotalAmount, Status)
VALUES
(1, '2024-01-10', 1499.00, 'Completed'),
(2, '2024-02-14', 599.00, 'Completed'),
(3, '2024-03-18', 3499.00, 'Completed'),
(4, '2024-04-20', 199.00, 'Completed'),
(5, '2024-05-28', 899.00, 'Completed'),
(6, '2024-06-22', 1299.00, 'Completed'),
(7, '2024-07-29', 799.00, 'Completed'),
(8, '2024-08-10', 2499.00, 'Completed'),
(9, '2024-09-16', 199.00, 'Completed'),
(10,'2024-10-19', 499.00, 'Completed'),
(11,'2024-11-05', 3499.00, 'Completed');


--TO find the total number of orderscount.
SELECT 'Orders'    AS TableName, COUNT(*) AS Rows FROM Orders;


select*from Orders
go


CREATE TABLE OrderItems
(
   OrderItemID                  INT                          IDENTITY(1,1) PRIMARY KEY,
```

```
    OrderID                 INT             FOREIGN KEY REFERENCES Orders(OrderID),
    ProductID               INT             FOREIGN KEY REFERENCES Products(ProductID),
    Quantity                INT,
    Price                   DECIMAL(10,2)

);

INSERT INTO OrderItems (OrderID, ProductID, Quantity, Price)
VALUES
(1, 1, 1, 1499.00),
(2, 2, 1, 599.00),
(3, 3, 1, 3499.00),
(4, 4, 1, 199.00),
(5, 6, 1, 899.00),
(6, 7, 1, 1299.00),
(7, 8, 1, 799.00),
(8, 9, 1, 2499.00),
(9, 10, 1, 199.00),
(10, 11, 1, 499.00),
(11, 3, 1, 3499.00);

--TO find the Total Orderitems count.
SELECT 'OrderItems'AS TableName, COUNT(*) AS Rows FROM OrderItems;

select*from OrderItems
```

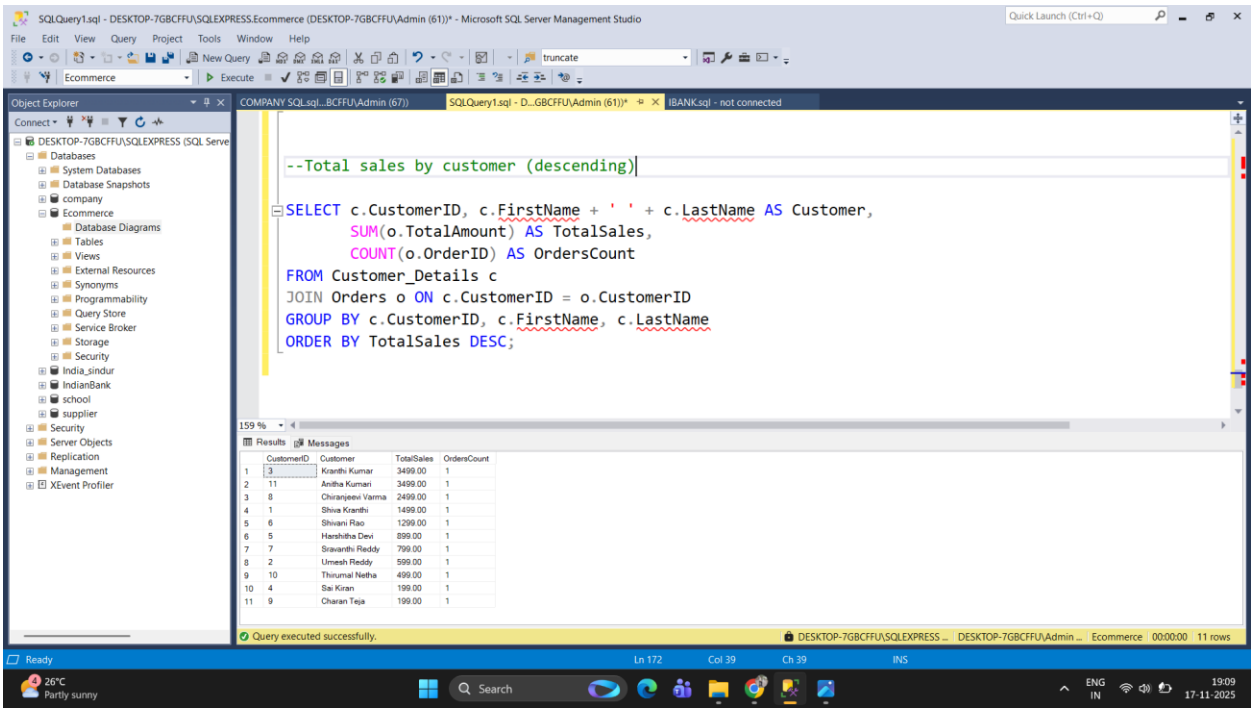
```
go


CREATE TABLE Payments (
   PaymentID          INT                    IDENTITY(1,1) PRIMARY KEY,
   OrderID            INT          FOREIGN KEY (OrderID) REFERENCES Orders(OrderID),
   PaymentDate        DATE,
   PaymentMethod      VARCHAR(20),
   Amount             DECIMAL(10,2)

);


INSERT INTO Payments (OrderID, PaymentDate, PaymentMethod, Amount)
VALUES
(1, '2024-01-10', 'UPI', 1499.00),
(2, '2024-02-14', 'Card', 599.00),
(3, '2024-03-18', 'UPI', 3499.00),
(4, '2024-04-20', 'Wallet', 199.00),
(5, '2024-05-28', 'Card', 899.00),
(6, '2024-06-22', 'UPI', 1299.00),
(7, '2024-07-29', 'UPI', 799.00),
(8, '2024-08-10', 'Card', 2499.00),
(9, '2024-09-16', 'UPI', 199.00),
(10,'2024-10-19', 'Wallet', 499.00),
(11,'2024-11-05', 'Card', 3499.00);
```

```
SELECT 'Payments'  AS TableName, COUNT(*) AS Rows FROM Payments;


select*from Payments

go


--Orders with customer and payment info

SELECT o.OrderID, o.OrderDate, c.FirstName + ' ' + c.LastName AS Customer,

      o.TotalAmount, o.Status,

      p.PaymentMethod, p.PaymentDate

FROM Orders o

JOIN Customer_Details c ON o.CustomerID = c.CustomerID

LEFT JOIN Payments p ON o.OrderID = p.OrderID

ORDER BY o.OrderDate;
```



--Total sales by customer (descending)

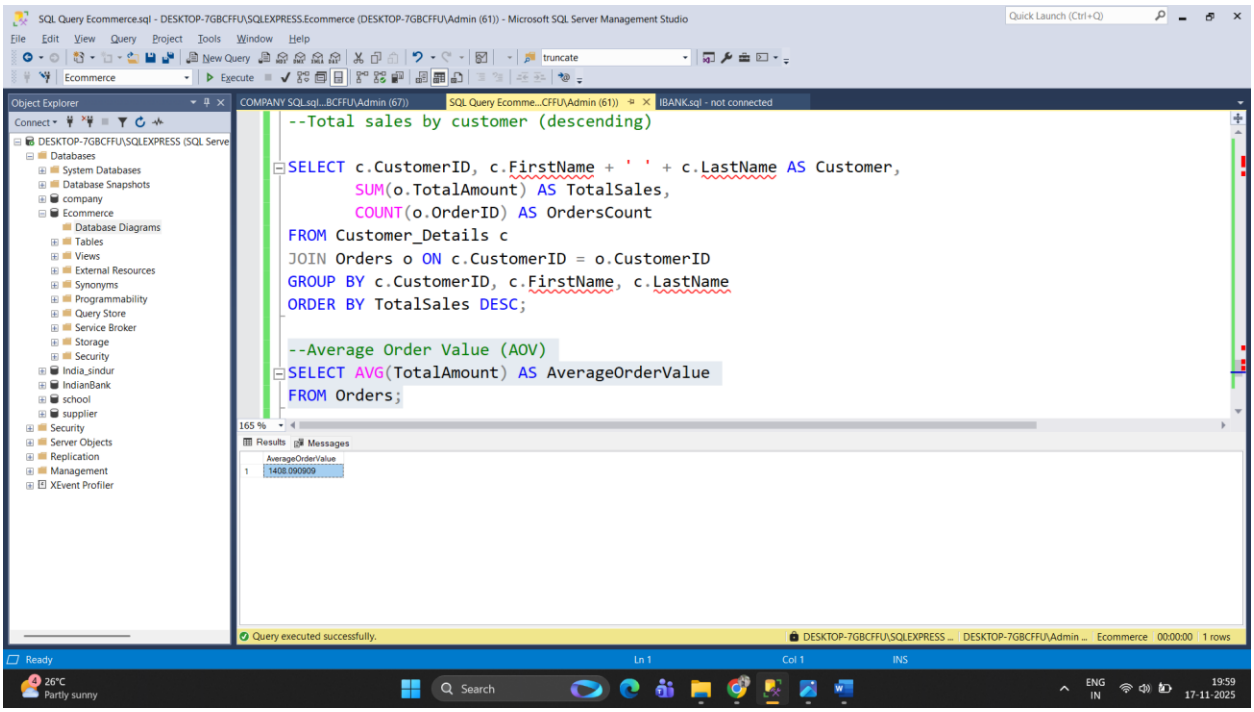
```
SELECT c.CustomerID, c.FirstName + ' ' + c.LastName AS Customer,
    SUM(o.TotalAmount) AS TotalSales,
    COUNT(o.OrderID) AS OrdersCount
FROM Customer_Details c
JOIN Orders o ON c.CustomerID = o.CustomerID
GROUP BY c.CustomerID, c.FirstName, c.LastName
ORDER BY TotalSales DESC;
```



--Average Order Value (AOV)

SELECT AVG(TotalAmount) AS AverageOrderValue

FROM Orders;



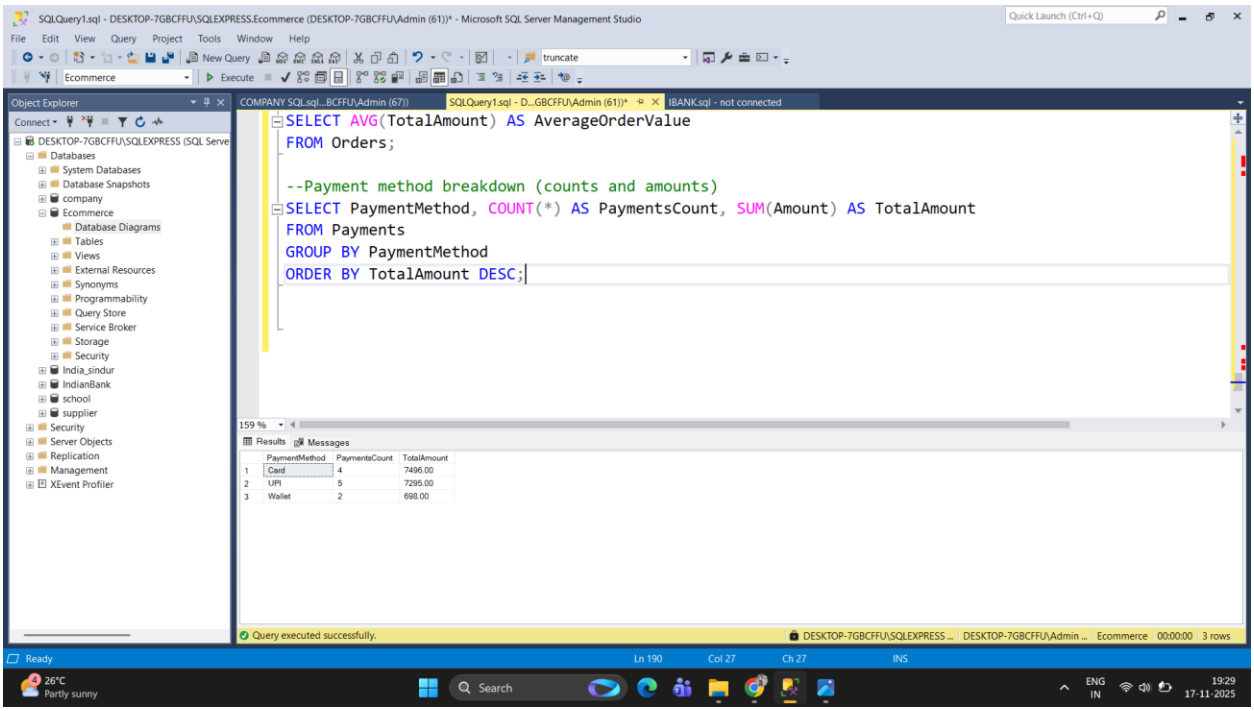
--Payment method breakdown (counts and amounts)

SELECT PaymentMethod, COUNT(*) AS PaymentsCount, SUM(Amount) AS TotalAmount

FROM Payments

GROUP BY PaymentMethod

ORDER BY TotalAmount DESC;


SQLQuery1.sql - DESKTOP-7GBCFFU\SQLEXPRESS.Ecommerce (DESKTOP-7GBCFFU\Admin (61))* - Microsoft SQL Server Management Studio
SELECT AVG(TotalAmount) AS AverageOrderValue
FROM Orders;
--Payment method breakdown (counts and amounts)
SELECT PaymentMethod, COUNT(*) AS PaymentsCount, SUM(Amount) AS TotalAmount
FROM Payments
GROUP BY PaymentMethod
ORDER BY TotalAmount DESC;
Results
Messages
PaymentMethod PaymentsCount TotalAmount
1 Card 4 7496.00
2 UPI 5 7295.00
3 Wallet 2 698.00
Query executed successfully.
Ready